

- 4 (a) When electrons interact with the metal target atoms, they decelerate and can lose their kinetic energies in various amounts. When an electron loses all its kinetic energy upon a single collision with a target atom, a photon (X-ray) is emitted. The energy of the photon emitted is a maximum value, which corresponds to a minimum wavelength.

(b) Using $E = \frac{hc}{\lambda}$, we have

$$E = \frac{6.63 \times 10^{-34} \times 3.0 \times 10^8}{1.54 \times 10^{-10}}$$

$$E \approx 1.29 \times 10^{-15} \text{ J}$$

- 5 (a) Maximum binding energy per nucleon $\approx 8.8 \text{ MeV}$